Machine learning for predicting sediment particle size distributions

UE24CS352A Machine Learning · Mini-project 62 · Team B-16
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3,998	9	0.86 / 0.82	0.60 / 0.62
samples from 92 casts	input features	R^2 (d50 / σ), random split	R^2 (d50 / σ), unseen casts

1. Problem statement

Suspended mud particles clump into flocs and break apart within seconds. Their size sets how fast they settle, and no physical model predicts it well. Following Egan (Stanford CS229, 2019), we use regression to predict the **median particle diameter d50** and the **spread σ** of the size distribution from routine water measurements, scored by R^2 on held-out data.

2. Dataset

The paper's data are not public. We used a U.S. Geological Survey release from the same bay (doi:10.5066/P99Q4CHM).

Coverage	Vertical casts at a channel site (11–14 m deep) and a shallows site (2–5 m) in south San Francisco Bay, on four days in July 2020.
Instruments	Laser particle sizer (36 size classes, 1.2–459 µm) and a CTD (salinity, turbidity, depth).
Size	193 raw files; after cleaning, 3,998 ten-second samples from 92 casts.
Not available	Chlorophyll, refractive index, wave and tidal velocity (used in the paper).

3. Approach

- **Features (9).** Depth, height above bed, water depth (tide stage), tide rate, temperature, salinity, turbidity, beam attenuation, site.
- **Models.** Random Forest and SVR as in the paper, plus ridge regression, gradient boosting and a small neural network. Tuned as in the paper.
- **Evaluation, our main extension.** Neighbouring samples are nearly identical, so a random split flatters the model. We test four ways: random 70/30 (the paper's), unseen casts, an unseen day, and an unseen site.

4. Implementation

build_dataset.py	Reads the raw files, aligns the two instruments' clocks, removes bad samples, averages into 10-second bursts, computes d50 and σ.
train_evaluate.py	Feature importance, tuning, five models, four test protocols.
demo.py	Holds out one cast, retrains, compares predicted with measured.

5. Results

R^2 on test data. Higher is better; below 0 is worse than predicting the mean. Shaded: the paper's protocol.

Model	Random 70/30 (paper)		Unseen casts		Unseen day		Unseen site	
	d50	σ	d50	σ	d50	σ	d50	σ
Ridge (linear baseline)	0.43	0.53	0.36	0.45	0.19	0.31	-5.39	< −10
Random Forest	0.86	0.82	0.56	0.58	0.35	0.46	-1.67	-0.95
SVR (RBF)	0.69	0.72	0.60	0.62	0.45	0.53	-1.90	-3.83
Gradient Boosting	0.85	0.81	0.59	0.60	0.42	0.50	-2.32	-1.17
Neural net (10 units)	0.69	0.72	0.53	0.62	0.34	0.52	< −10	-7.09

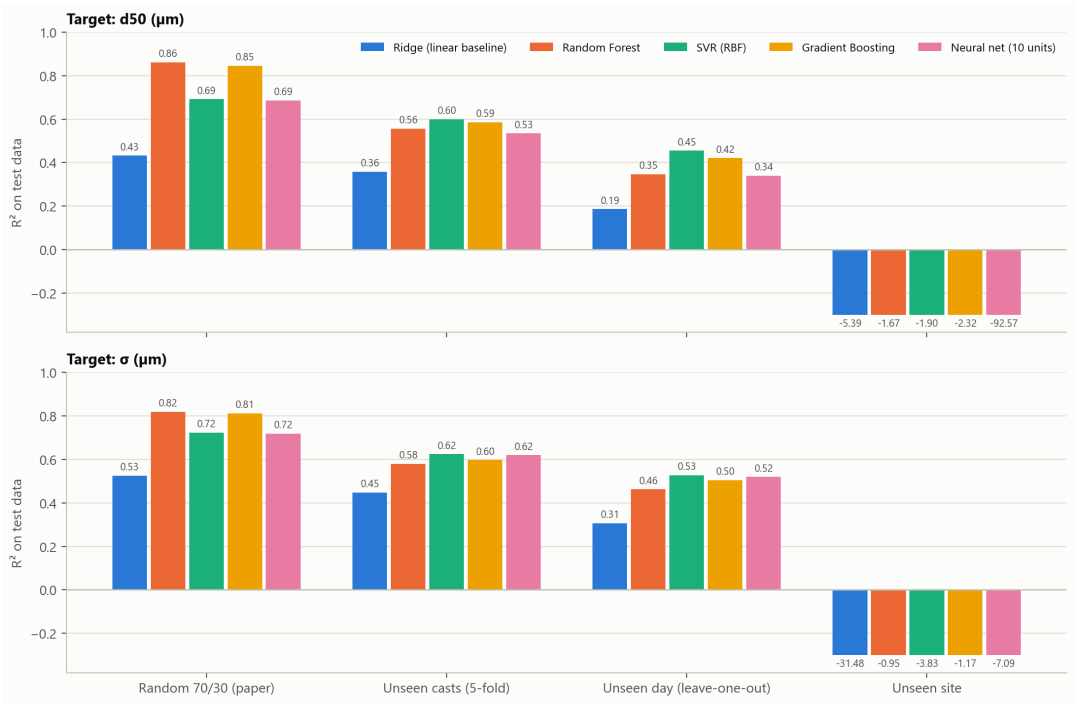


Figure 1. R^2 of every model under the four test protocols (bars cut off at −0.3).

- **The paper's result reproduces.** Random split, Random Forest: R^2 = 0.86 (d50) and 0.82 (σ). The paper reports 0.81 and 0.75.
- **Part of that is memorised neighbours.** On unseen casts the best model (SVR) scores 0.60 and 0.62.
- **Nothing transfers between sites.** Every model scores below zero.
- **Salinity matters little here.** It carries 8 % of the importance, against 72 % in the paper, because it only ranged from 30.1 to 32.3 psu. Depth, tide stage and beam attenuation matter most.

6. Conclusions

Machine learning explains about 60 % of the variation in floc size on new casts: useful, but well short of what a random split suggests. For field data like this the split decides the score, so the paper's worry about overfitting was justified. Limits: four days, two sites, a narrow salinity range. Next: add current and wave data and more sites.